SMART POWER DISTRIBUTION MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

GUGANRAJ R-8115U25CS050

in partial fulfillment of requirements for the award of the course

CGB1122 – DATA STRUCTURES

in

COMPUTER SCIENCE AND ENGINEERING

Under the guidance of

Mrs.R.NALINI

COMPUTER SCIENCE AND ENGINEERING

K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY 621112

ANNA UNIVERSITY

CHENNAI 600025

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(AUTONOMOUS)
SAMAYAPURAM – 621 112

BONAFIDE CERTIFICATE

Certified that this project report titled **"SMART POWER DISTRIBUTION MANAGEMENT SYSTEM**" is the Bonafide work of **GUGANRAJ R(8115U25CS050)**, who carried out the project work under my supervision.

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Submitted for the end semester examination held on ……

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Under

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DECLARATION BY THE CANDIDATE

I jointly declare that the project report on **"INTEGRATED ONLINE MEDICINE ODERING AND HEALTHCARE HUB BY USING PHP (MEDILINK)"** is the result of original work done by us and best of our knowledge. This project report is submitted on the partial fulfillment of the requirement of the award of the course **CGA1121- DATA STRUCTURES**

Signature

GUGANRAJ R

Place: Samayapuram

Date:

ACKNOWLEDGEMENT

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I render our sincere thanks to the Course Coordinator and other staff members for providing valuable information during the course.

I wish to express our special thanks to the officials and Lab Technicians of our departments who rendered their help during the period ofthe work progress.

INSTITUTE VISION AND MISSION

VISION OF THE INSTITUTE:

To achieve a prominent position among the top technical institutions.

MISSION OF THE INSTIITUTE:

M1: To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.

M2: To nurture research and entrepreneurial skills among students in cutting edge technologies.

M3: To provide education for developing high-quality professionals to transform the society.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations

6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The Smart Power Distribution Management System is a Data Structures based project designed to improve the efficiency, reliability, and automation of electrical power distribution. This system manages consumer information, monitors power load distribution, handles fault management, and generates billing reports in an organized manner. The project uses Linked List for consumer management, Binary Search Tree (BST) for fast consumer lookup, and Queue for fault handling using First Come First Serve (FCFS) technique. It also includes load balancing analysis and power station monitoring to ensure stable power distribution. This system helps reduce manual work, improve fault response, optimize energy management, and provide a smarter approach to modern power distribution network.

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LIST OF ABBREVIATIONS

ABBREVIATIONS

SLL - Singly Linked List

UI - User Interface

FCFS - First Come First Serve

BST - Binary Search Tree

MAX - Maximum Queue Size

DBMS - Database Management System

kWh - Kilowatt-hour

GIS - Geographic Information System

NULL - Null Value

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The Smart Power Distribution Management System is designed to automate and improve the management of electrical power distribution. Traditional power distribution systems often face problems such as inefficient load management, delayed fault handling, and manual consumer monitoring. This project provides a smart solution to overcome these challenges using Data Structures concepts.

1.2 PURPOSE AND IMPORTANCE

The Smart Power Distribution Management System is developed to automate, monitor, and optimize the process of electrical power distribution using Data Structures. The main purpose of this project is to manage consumer records efficiently, monitor power load distribution, handle fault detection and resolution, perform quick consumer lookup, and generate accurate billing reports. It aims to reduce manual intervention, improve operational efficiency, and ensure reliable and uninterrupted power supply through smart management techniques.

This project is important because it helps achieve balanced load distribution, reduces overload conditions and power losses, and improves fault handling through faster response mechanisms. It enhances the efficiency of consumer management and simplifies monitoring of power stations and substations. By automating important processes such as fault control, load balancing, and billing, the system improves reliability, stability, and overall performance of the power distribution network.

In addition, the system supports better decision-making in power management by providing organized data handling and real-time monitoring capabilities. It contributes to smarter energy utilization, improved service quality for consumers, and faster maintenance response. The project also demonstrates the practical implementation of Data Structures such as Linked List, Binary Search Tree, and Queue in solving real-world problems related to smart grid technology, making it both academically valuable and industrially relevant.

1.3 OBJECTIVES

1. Manage consumer records.
2. Monitor power distribution.
3. Handle faults quickly.
4. Search consumer details fast.
5. Generate billing reports.

1.4 PROJECT SUMMARIZATION

The Smart Power Distribution Management System is developed to automate and improve power distribution management using Data Structures. The project manages consumer records, monitors load distribution, handles faults, performs consumer lookup, and generates billing reports efficiently. It uses Linked List for consumer management, Binary Search Tree for searching, and Queue for fault handling.

The project also helps maintain balanced power supply, supports faster fault resolution, and improves overall system performance through automation. By integrating multiple data structures into a single application, the system provides a smart and organized solution for modern power distribution management and shows how programming concepts can be applied to solve practical engineering problems And some benefitsof this project are as follows:-

Dynamic and Flexible:- The system is dynamic because it can handle changing consumer data, power loads, and faults efficiently.

User-Friendly:- The system is user friendly because it provides a simple interface to manage consumers, monitor power distribution and handle faults.

Data Integrity:- The system maintains data integrity by storing consumer and power distribution data accurately and consistently.

Persistence:- The system provides persistence by maintaining and preserving consumer and power distribution data for continuous use

CHAPTER 2

PROJECT METHODOLOGY

2.1 INTRODUCTION TO SYSTEM ARCHITECTURE

The System Architecture of the Smart Power Distribution Management System represents the overall structure and working flow of the project. It shows how different modules such as Consumer Control, Power Station Dashboard, Load Distribution Monitor, Fault Control Center, Billing System, and Consumer Search are interconnected and function together within the system.

2.1.1 High-Level System Architecture

The high-level system architecture for the phone directory application typically consists of several key components:

(i) User Interface (UI)

(ii) Application Logic

(iii) Data Management Layer

2.1.2 Components of the System Architecture

a. User Interface (UI)

The User Interface (UI) is the interaction layer between the user and the Smart Power Distribution Management System. It allows users to access the system, enter commands, and perform operations through the main menu. Through the UI, users can manage consumers, monitor load distribution, handle faults, generate billing reports, and search consumer details easily.

b. Application Logic

The Application Logic controls the core functioning of the Smart Power Distribution Management System. It processes user inputs, executes operations, and manages the working of modules such as consumer management, load balancing, fault handling, billing, and consumer search.

c. Data Management Layer

The Data Management Layer is responsible for managing the data structures used to store and retrieve contact information efficiently.

2.2 DETAILED SYSTEM ARCHITECTURE DIAGRAM

It illustrates how the main menu connects with different functional modules such as Consumer Control, Power Station Dashboard, Load Distribution Monitor, Fault Control Center, Billing System, and Consumer Search.

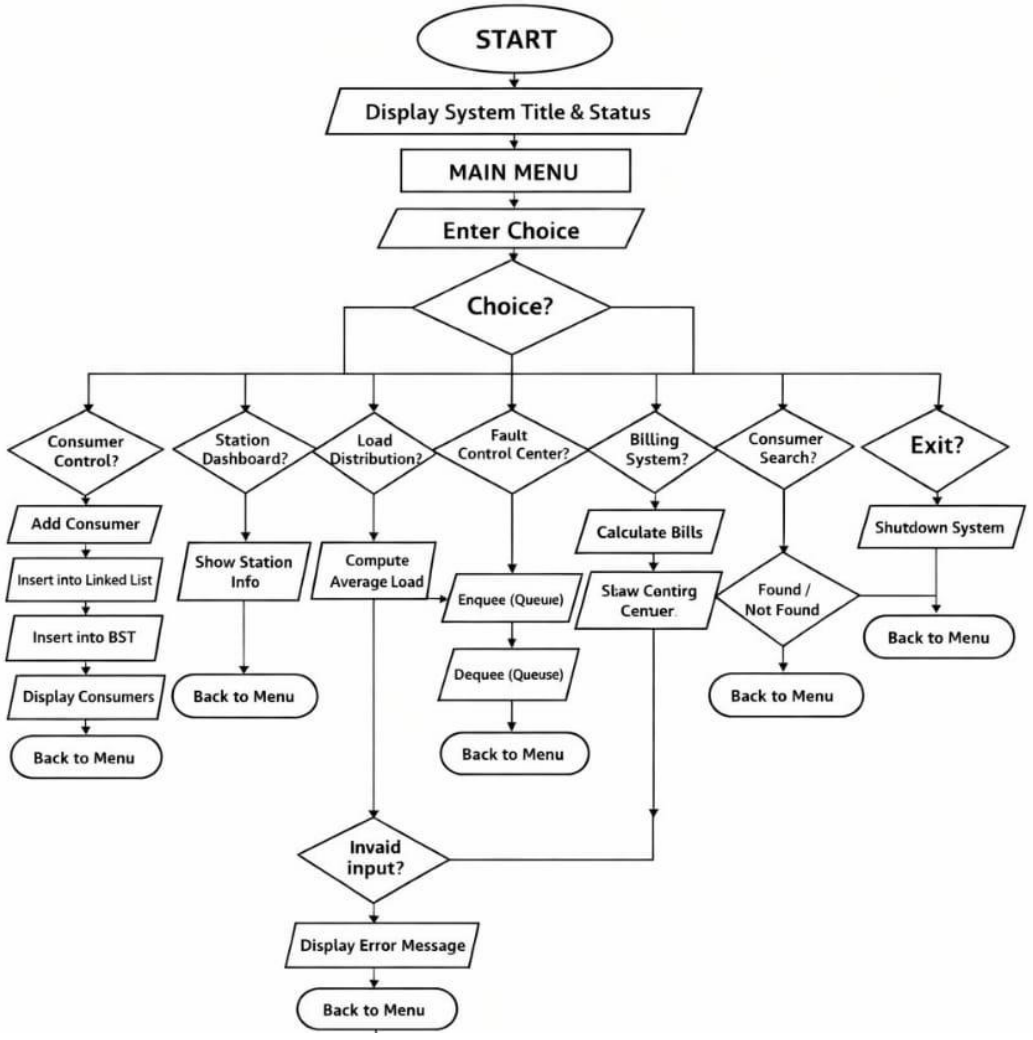


Figure 2.1 : Architecture Diagram (Sample)

CHAPTER 3

DATA STRUCTURE PREFERANCE

3.1 EXPLANATION OF WHY A SINGLY LINKED LIST WAS CHOSEN

Explanation of Why a Linked List Was Chosen A Linked List was chosen because it allows dynamic storage and management of consumer records efficiently. In the Smart Power Distribution Management System, the number of consumers can increase or change at any time, and a linked list makes it easy to add new consumers without fixed memory limitations.

Dynamic Memory: Linked List uses dynamic memory allocation, which allows memory to be assigned when needed. It does not require a fixed size for storing data. This helps the system handle increasing consumer records easily. It improves memory usage and supports efficient data storage.

Flexible Data Handling: Linked List can handle updates and changes in data efficiently. Records can be added or modified easily when needed. This makes the system flexible for changing consumer information. It supports smooth data management.

Easy Insertion: New consumer records can be added easily in a linked list without shifting existing data. Each new node can be connected directly to the list. This makes insertion faster and simpler. It is useful when consumer data changes frequently.

Managing Consumer Records: Linked List helps organize consumer details such as ID, name, location, and energy usage. Each consumer record is stored as a separate node. This makes managing and accessing records easier. It improves the organization of consumer data.

Efficient Storage: Linked List stores data using nodes connected through links. It uses memory as required instead of allocating unused space. This improves storage efficiency. It makes the structure suitable for managing consumer information.

3.2 COMPARISON WITH OTHER DATA STRUCTURES

List was chosen over other data structures because it provides dynamic memory allocation, easy insertion, and flexible data handling. Unlike arrays, it does not have fixed size limitations or require shifting data during insertion. Compared to stacks and queues, which follow specific data access orders, Linked List is more suitable for storing and managing multiple consumer records

3.3 ADVANTAGES AND DISADVANTAGES OF USING A DLL

3.3.1 Advantages of Using a Singly Linked List:

A Linked List offers several advantages in the Smart Power Distribution Management System. It supports dynamic memory allocation, allowing the system to handle increasing consumer records without fixed size limitations. It enables easy insertion of new consumer data without shifting existing records. Linked List provides flexible data handling, making updates and management simpler. It also improves storage efficiency by using memory only when required. These advantages make Linked List suitable for managing consumer information effectively in the project.

3.3.2 Disadvantages of Using a Singly Linked List:

A Linked List also has some disadvantages. It requires extra memory to store pointers for linking nodes. Accessing a specific record is slower because data is accessed sequentially. Searching can take more time compared to some other data structures. It is more complex to manage than arrays due to pointer handling. These limitations can affect performance in certain situations.

CHAPTER -4

DATA STRUCTURE METHODOLOGY

4.1 SINGLY LINKED LIST

A Singly Linked List is a linear data structure where each node contains data and a pointer to the next node. In this project, it is used to store and manage consumer records dynamically. Each consumer node contains details such as ID, name, location, energy usage, and a next pointer linking to the next consumer record. It allows easy insertion of new consumers, flexible data handling, and efficient management of changing consumer data.

Key features of a circular doubly linked list:

1.Dynamic Memory AllocationNodes: A Singly Linked List uses dynamic memory allocation, allowing memory to be assigned when needed. It does not require a fixed size and can handle increasing data efficiently.

2.Node-Based Sequential Structure: Each node contains data and a pointer to the next node, creating a sequential connection between records. This structure helps organize and manage data effectively.

3.Easy Insertion and Data Management: New nodes can be added easily without shifting existing data. This makes insertion simple and supports flexible management of changing records.

4.Efficient Handling of Dynamic Data: A Singly Linked List is suitable for applications where data changes frequently. It allows efficient handling of growing records and supports dynamic system operations.

4.2 NODE STRUCTURE

A node is the basic building block of a Singly Linked List. Each node contains two main parts: a data field and a pointer field. The data field stores consumer information such as ID, name, location, and energy usage. The pointer field, called next, stores the address of the next node in the list. This helps connect nodes in sequential order.

The typical structure of a node ina circular doubly linked list used for a Consumer Management might include the following components:

1. **Data Fields: -** A node in a linked list has two parts: **data field** and **pointer**. The data field stores the actual information like Consumer ID, name, location, and energy usage. It is used to keep the required details of each consumer in the system.
2. **Pointers: -** The pointer field stores the address of the next node. It helps connect one node to another in a sequence. This linking allows easy movement from one consumer record to the next in the system.

All nodes are connected in a sequential manner using the next pointer, forming a chain-like structure. The first node is accessed using a head pointer, and traversal continues until the last node, which points to NULL.This structure allows efficient management of consumer records in a dynamic way. It supports easy insertion of new consumers and helps maintain organized and linked data storage within the system

4.3. INITIALIZATION AND INSERTION

Initialization means starting the linked list by setting the head pointer to NULL, which shows the list is empty. When the first consumer is added, the head pointer points to that node.Insertion means adding a new consumer into the list. A new node is created, data is stored, and it is linked using the next pointer. It can be added easily at the beginning or end of the list

CHAPTER-5

MODULES

5.1 ADD A CONSUMER :

- **Function Name:** addConsumer()
- **Description:** The Consumer Management Module is used to handle all consumer-related operations in the Smart Power Distribution Management System. It allows the system to add new consumers, store their details such as ID, name, location, and energy usage, and manage them efficiently using a linked list.

5.2 POWER STATION DETAILS :

- **Function name**: stationDashboard()
- **Description:-** The Power Station Dashboard Module is used to display the overall status of the power distribution system. It shows important details such as station name, number of substations, current load condition, and system status. This helps in monitoring whether the power grid is running normally or not.

5.3 LOAD CHECKING :

- **Function name:** loadBalancing()
- **Description:-** This function calculates total load, average load, and monitors power distribution across consumers.

5.4 FAULT CONTROL

- **Function Name:** enqueue()dequeue()
- **Description:** The Fault Management Module is used to handle power fault complaints in the system. When a fault is reported, it is added to a queue using the enqueue operation.The faults are resolved one by one in First Come First Serve (FCFS) order using the dequeue operation. This ensures proper and fair handling of all fault requests in the system.

5.5 BILL REPORT :

- **Function name**: billing()
- **Description:-** This function calculates and displays the electricity bill based on units consumed by each consumer.

5.6 CONSUMER STATUS :

- **Function name:** searchBST()
- **Description:-** This function searches for a consumer using Consumer ID in the Binary Search Tree (BST) and displays their status.

5.7 SHUTDOWN :

- **Function name:** main()exit()
- **Description:-** The shutdown option is handled inside the main() function.It uses exit(0) to safely terminate the program and close the system

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

The Smart Power Distribution Management System successfully achieves efficient and reliable power distribution management using Data Structures. It integrates consumer management, load monitoring, fault handling, and billing into a single organized system. By using Linked List, Binary Search Tree, and Queue, the project improves automation, reduces manual work, and enhances system performance.

Overall, the project provides a smart approach to modern power distribution by ensuring better fault response, balanced load management, and efficient consumer data handling. It demonstrates the practical application of Data Structures in solving real-world problems and contributes to the development of reliable and intelligent power management systems.

6.2 FUTURE SCOPE

The Smart Power Distribution Management System can be further improved by adding advanced technologies such as Artificial Intelligence and Machine Learning for better load prediction and fault detection. It can also be connected with real-time sensors and IoT devices to monitor power distribution automatically. A web or mobile application interface can be developed to make the system more user-friendly and accessible from anywhere.

In the future, the system can be expanded to include smart energy optimization, renewable energy integration, and automated decision-making for power distribution. It can also support large-scale smart grid networks with cloud storage for better data management and analysis. These improvements will make the system more efficient, intelligent, and suitable for modern power management systems. Additionally, features such as automatic fault alerts, remote monitoring, and real-time consumer usage tracking can be added to improve system performance. The project can also be enhanced with cybersecurity measures to protect power distribution data and ensure secure system operation. With these developments, the system can evolve into a complete smart grid solution for future energy management.

APPENDICES

APPENDIX A-SOURCE CODE

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
/// ================= LINKED LIST ================
struct Consumer {
   int id;
   char name[50];
   char location[50];
   float units;
   struct Consumer* next;};
struct Consumer* head = NULL;
/// ================= BST ================
struct BST {
   int id;
   struct BST *left, *right;};
struct BST* root = NULL;
/// ================= QUEUE ================
#define MAX 10
int queue[MAX], front = -1, rear = -1;
/// ================= BST FUNCTIONS ================
struct BST* insertBST(struct BST* root, int id) {
   if (!root) {
      struct BST* newNode = malloc(sizeof(struct BST));
      newNode->id = id;
      newNode->left = newNode->right = NULL;
      return newNode}
   if (id < root->id)
      root->left = insertBST(root->left, id);
   else
      root->right = insertBST(root->right, id);
   return root;}
void searchBST(struct BST* root, int id) {
   if (!root) {
      printf("\n ✗ Consumer Not Found!\n");
      return; }


   if (root->id == id) {
      printf("\n✓ Consumer Found in Network\n");
 printf("Status: ACTIVE | Supply: NORMAL\n");
```

```
        return; }
    if (id < root->id)
        searchBST(root->left, id);
    else
        searchBST(root->right, id);}
/// ================ CONSUMER MANAGEMENT ================
void addConsumer() {
    struct Consumer* newNode = malloc(sizeof(struct Consumer));
printf("\n➤ Registering New Consumer...\n\n");
    printf("Consumer ID      : ");
scanf("%d", &newNode->id);
printf("Name             : ");
    scanf("%s", newNode->name);
  printf("Location         : ");
    scanf("%s", newNode->location);
printf("Energy Usage     : ");
    scanf("%f", &newNode->units);
newNode->next = head;
    head = newNode;
 root = insertBST(root, newNode->id);
printf("\n✓ Consumer Successfully Integrated into Grid\n");}
void displayConsumers() {
    struct Consumer* temp = head;
 printf("\n[ACTIVE CONSUMERS]\n");
    printf("-------------------------------------------------\n");
while (temp != NULL) {
        printf("ID: %d | Name: %-6s | Area: %-10s | Load: %.0f Units\n",
               temp->id, temp->name, temp->location, temp->units);
        temp = temp->next; }
printf("-------------------------------------------------\n");}
/// ================ LOAD BALANCING ================
void loadBalancing() {
    struct Consumer* temp = head;
    float total = 0;
    int count = 0;
while (temp) {
        total += temp->units;
        count++;
        temp = temp->next;  }


printf("\n[LOAD DISTRIBUTION MONITOR]\n");
    printf("-------------------------------------------------\n");
```

```
 if (count == 0) {
        printf("No Data Available\n");
        return;    }
printf("Total Consumers      : %d\n", count);
    printf("Total Load Demand    : %.0f Units\n", total);
    printf("Average Load         : %.0f Units\n", total / count);
 printf("\n✓ Load Balanced Across Substations\n");
    printf("✓ No Overload Detected\n");
    printf("-------------------------------------------------\n");}
/// ================ FAULT MANAGEMENT ================
void enqueue(int id) {
    if (rear == MAX - 1) {
        printf("Queue Full!\n");return;   } if (front == -1) front = 0;
queue[++rear] = id;
    printf("\n⚠ Fault Logged Successfully → Sent to Service Queue\n");}

void dequeue() {
    if (front == -1 || front > rear) {
        printf("No Faults!\n");
        return;    }
    printf("\n🔧 Resolving Fault ID: %d...\n", queue[front++]);
    printf("✓ Power Restored Successfully\n");}
/// ================ BILLING ================
void billing() {
    struct Consumer* temp = head;
 printf("\n[ENERGY BILLING REPORT]\n");
    printf("-------------------------------------------------\n");
while (temp) {
        printf("ID: %d | %-6s | Units: %.0f | Bill: ₹%.0f\n",
              temp->id, temp->name, temp->units, temp->units * 5);
        temp = temp->next;  }
 printf("\n✓ Reports Generated Successfully\n");
    printf("-------------------------------------------------\n");}
/// ================ STATION ================
void stationDashboard() {
    printf("\n[POWER STATION DASHBOARD]\n");
    printf("-------------------------------------------------\n");
    printf("Station Name     : Chennai Main Grid\n");
    printf("Substations      : 3 Active\n");

    printf("Current Load     : Stable\n");
    printf("\n✓ Power Distribution Running Normally\n");
    printf("-------------------------------------------------\n");}
```

```
/// ================ MAIN ================
int main() { int choice, sub, id;
printf("=================================================
\n");
printf("   ⚡ TNEB SMART POWER DISTRIBUTION SYSTEM ⚡\n");

printf("=================================================
\n");
   printf("Status        : ONLINE\n");
   printf("Region        : Tamil Nadu Grid\n");
   printf("Control Mode  : Automatic\n");

printf("=================================================
\n");
while (1) {
       printf("\nMAIN MENU\n");
       printf("-------------------------------------------------\n");
       printf("1. 👨‍💼 Consumer Control Panel\n");
       printf("2. 🏭 Power Station Dashboard\n");
       printf("3. ⚖️ Load Distribution Monitor\n");
       printf("4. 🚨 Fault Control Center\n");
       printf("5. 💰 Billing & Energy Report\n");
       printf("6. 🔍 Consumer Lookup\n");
       printf("7. ❌ Shutdown System\n");
       printf("-------------------------------------------------\n");
 printf("Enter Command: ");
       scanf("%d", &choice);
 switch (choice) {
          case 1:
             addConsumer();
             displayConsumers();
             break;
case 2:
             stationDashboard();
             break;
case 3:
             loadBalancing();
             break;


case 4:
             printf("\n[FAULT CONTROL CENTER]\n");
             printf("1. Report Fault\n2. Resolve Fault\n");
```

```
            printf("Enter Option: ");
            scanf("%d", &sub);
 if (sub == 1) {
               printf("\nFault ID        : ");
               scanf("%d", &id);
               printf("Fault Location  : Line\n");
               enqueue(id);
            } else {
               dequeue();
            }
            break;
 case 5:
            billing();
            break;
case 6:
            printf("\n[CONSUMER SEARCH]\n");
            printf("Enter Consumer ID: ");
            scanf("%d", &id);
            searchBST(root, id);
            break;
case 7:
            printf("\n⚡ System Shutting Down...\n");
            printf("✓ Thank You for Using Smart Grid System\n");
            exit(0);
default:
            printf("Invalid Command!\n");   }   }}
```

APPENDICES

APPENDIX B - SCREENSHOTS

1.ADD A CONSUMER :

```
==================================================
    ⚡ SMART POWER DISTRIBUTION MANAGEMENT SYSTEM ⚡
==================================================
Status         : ONLINE
Region         : Tamil Nadu Grid
Control Mode   : Automatic
==================================================

MAIN MENU
--------------------------------------------------
1. 👤Consumer Control Panel
2. 🏭Power Station Dashboard
3. ⚖️Load Distribution Monitor
4. 🚨Fault Control Center
5. 💰Billing & Energy Report
6. 🔍Consumer Status
7. ❌Shutdown System
--------------------------------------------------
Enter Command: 1

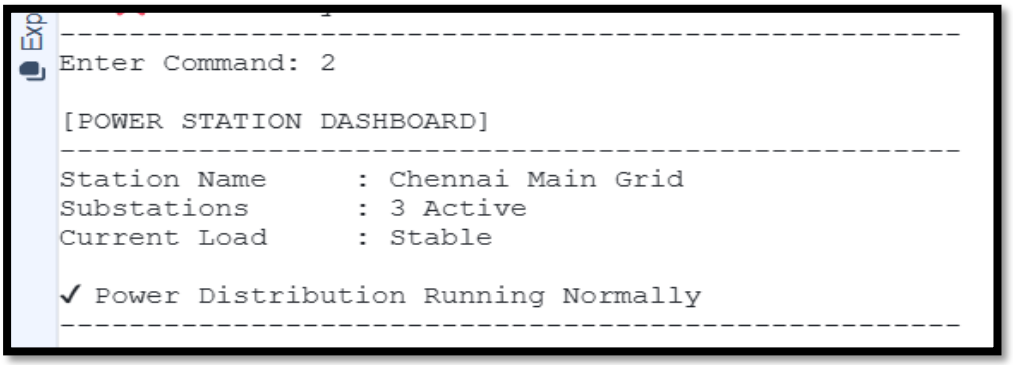
➤ Registering New Consumer...

Consumer ID       : 2008
Name              : Gugan
Location          : Trichy
Energy Usage      : 1000

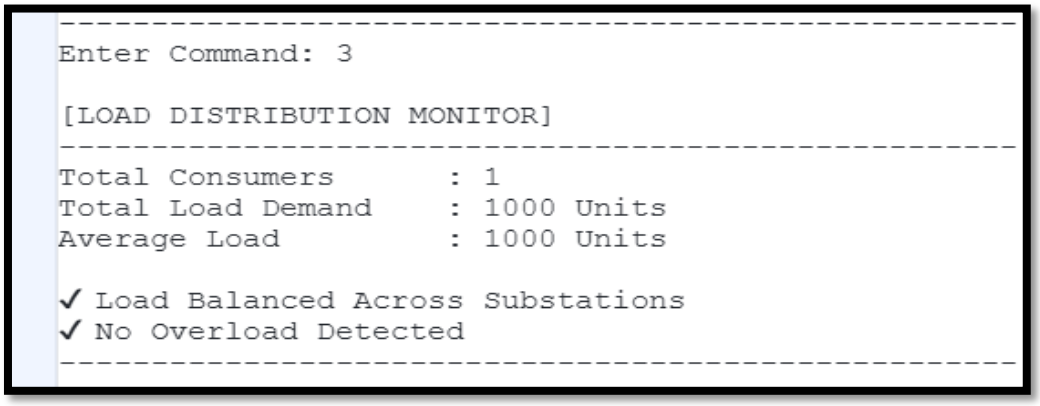
✓ Consumer Successfully Integrated into Grid

[ACTIVE CONSUMERS]
--------------------------------------------------
ID: 2008 | Name: Gugan  | Area: Trichy     | Load: 1000 Units
--------------------------------------------------
```

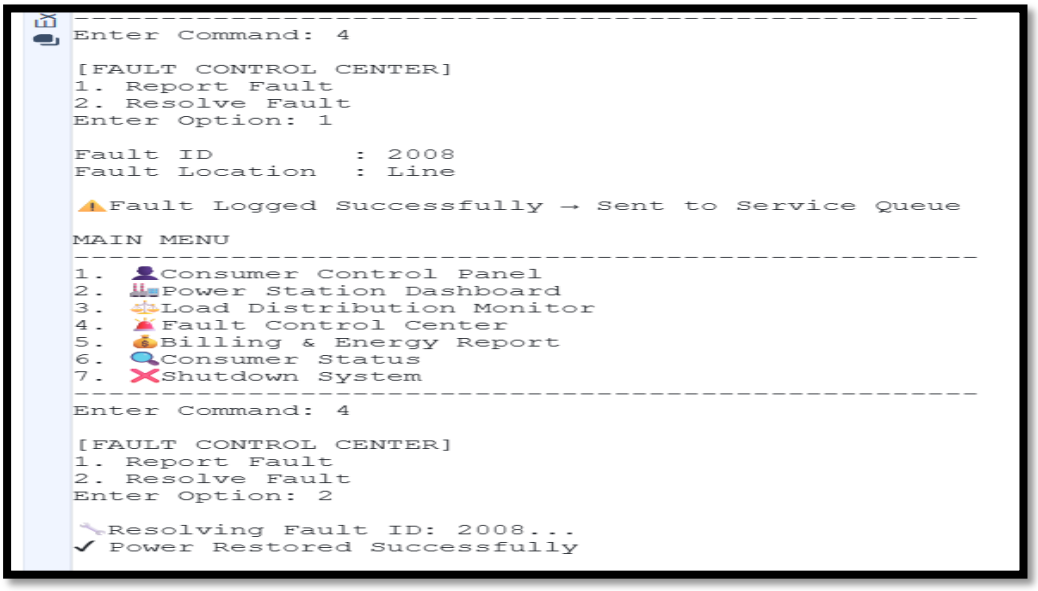
2.POWER STATION DETAILS :



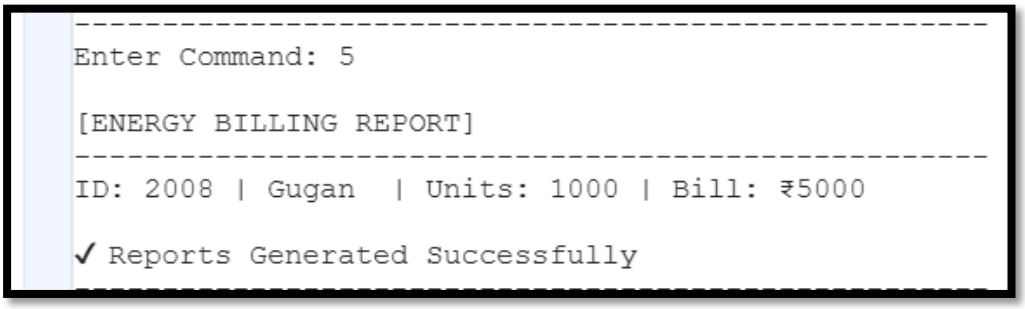
3.LOAD CHECKING :



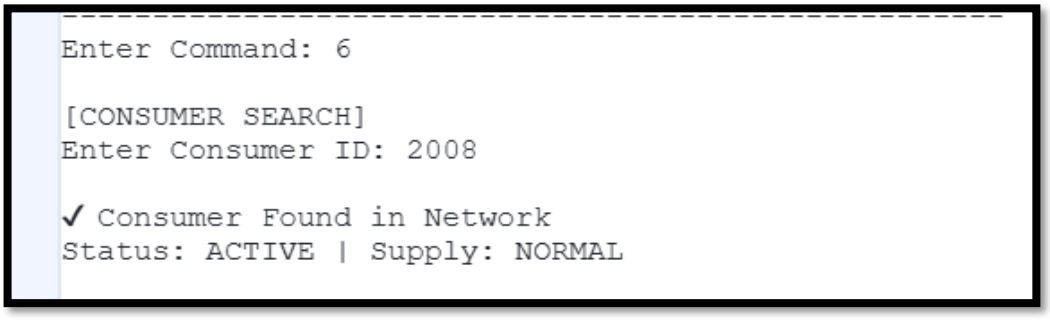
4.FAULT CONTROL :



5.BILL REPORT :



6.CONSUMER STATUS :



7.SHUTDOWN :

